Causal Attitude Networks on the Substrate

What Moved, What Did Not, and What the Mean Field Costs

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DRAFT — draft-2026-09-02.1 r3 (2 September 2026). Circulated for comment, not as a finished result. **This revision reports the programme's record to 14 August 2026**; its first revision reported the same record as it stood on the morning of 13 August, and the record moved past all three of that revision's findings within two days. Every remark below therefore carries the date of the rung it reports. Two of the first revision's five falsifiability commitments have fired against the record; both are reported as fired. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

The Causal Attitude Network (Dalege et al., 2016) models an attitude as an Ising-family network of interacting evaluative components. *The headline is that the model was moved from a hosted solver to native graph rules, and that the audit of that move corrected two of this paper's own earlier findings.* It was reimplemented on this substrate *twice*: first as a hosted mean-field self-consistency solver running beside the graph, then — one day later — as graph rules over graph data, in the existing vocabulary and with no new primitive in the compiled runtime. Five findings carry that, and two of them correct this paper's own earlier revision. The *alignment* claim, that the dynamics had no term surface, has failed: they have one, and the remaining boundary is consumer-side and about scope. The two-cycle oscillation reported as a property of the borrowed mathematics was a property of the update schedule, and vanishes on every one of 126 re-tested cells under damping. The *performance* question is still not the complexity one it appears to be, and the ladder above mean field has now been climbed: approximate propagation in both variants, with a two-sided result and a 20–200× cost, and exact inference where the treewidth permits it. Of two registered predictions, one is confirmed with a large effect and a control, and one is *reversed* — and survives its own solver correction in direction while losing it in magnitude. All of this is reported as measured rather than reinterpreted.

1 What was moved, what a move would mean, and on what standard

An attitude in the CAN model is a network of evaluative components with pairwise couplings and external fields, and its behaviour is the behaviour of an Ising system: relaxation toward low-energy configurations, frustration where couplings conflict, and a temperature-like parameter governing how sharply it settles.

Moving a formalism onto this substrate means more than making it run there. The substrate's claim — the substrate is the one set out in *The Law of Graphic Equalisation — Theory* (referred to below as Graphic Equalisation) — is that decisions are taken by graph rules over graph data; a model that computes its result in a compiled routine and hands back an answer has been

hosted, not moved. The distinction is the whole of the alignment question below, and it is not rhetorical: the two cases differ in what can subsequently be inspected, mutated, minted and reasoned over by the rest of the account. It is also the criterion that decides Remark 2.2, and the criterion was met after this paper's first revision said it had not been.

The standard, stated once. Every figure below is a *measurement* reported from this programme's own record rather than from a published source. What is offered is a pre-registered implementation record which is not circulated with these papers, and it is a claim a reader may decline to credit on exactly the terms Graphic Equalisation sets for the same case. Each rung of that record fixes its own commit, seed and sample before it runs, and reports what was compared, what was found, and the scope the run holds under; it names no implementation identifiers, and neither does this paper. The standard is not re-disclaimed at each remark. What each remark does carry is the *date* of the rung it reports, because the defect this revision repairs was a dating one.

2 Alignment: reimplemented twice

The model arrived as a hosted solver: a compiled mean-field self-consistency routine, exposed to the host and driven by seed-resident network data. The rules written against it did nothing, and the reason was worth having.

Remark 2.1 (The hosted stage, and the null that dated within a day). *(Record of 13 August 2026.)* There was no term surface for the dynamics at all, so the fifteen rules written against the network were inert — and the obvious diagnosis, missing or malformed match clauses, was not the reason. The relaxation existed only across a host-language boundary, so a rule could not express it, could not be matched against it, and could not mint against it. The rules were not broken; there was nothing for them to be written in terms of. Exactly one of the fifteen described the *recognition* half rather than the update half, and that one was wired with real match, where and effects clauses using only the existing term vocabulary; it works on the production path. *So the boundary was not a wall*, and the narrow statement that followed — what had moved was the network's data and its results, not its dynamics — was true for about one day.

Remark 2.2 (The dynamics are native, and two engine findings came with them). *(Record of 13 August 2026.)* The update was then expressed in the existing term vocabulary as five rules and four admissibility productions, with *no new primitive added to the compiled runtime*: eight tests, three of them live demonstrations with measured numbers. A concept's local field is the sum of its two field attributes and the coupling-weighted states of its neighbours; a single rule commits the thresholded value of that field — three-valued, with a neutral band — whenever it differs from the state held. The update is asynchronous by the engine's own execution model rather than by arrangement: one binding is applied and committed per pass, and the next match reads the graph as it now stands. Across twenty draws per condition it failed to converge on *none*, in either vocabulary and under either state alphabet, where the continuous synchronous relaxation oscillated on 12/20 coarse and 4/20 fine draws. For the two-valued case the absence of a cycle is structural rather than observed — every committed flip strictly decreases a bounded energy — and for the three-valued case it is an empirical fact on these draws and is reported as one. This is the criterion Section 1 gives for *moved* rather than *hosted*, and it is met.

Two engine-level findings were made and corrected while building, and both belong to the machinery rather than to the model. *First*: where sibling rules compete to update the same kind of node, the pass scheduler lets the first with a replenished candidate pool take every pass and starve the others indefinitely. It presented as non-convergence and was diagnosed as its opposite — nodes frozen at their initial value, the system already at a reachable fixed point it could not reach — and was fixed by collapsing the sign branches into one rule that computes its

target. *Second*: the decoder's search prefers a redundant composite explanation to a clean single exact match at equal coverage, because a coverage tie is broken on accumulated fit rather than on fewest parts. That is an Occam violation in a tie-break; it was worked around by scoping the codebook to one candidate at a time, and is reported rather than fixed.

What still runs compiled, and the one reason. Of the six consumers remaining after that build, two were migrated and four were not — and the four share *one* confirmed cause, not the three separate ones an earlier audit had named. The rule engine has no per-network scope: firing it on behalf of one network advances every other network in the graph, confirmed directly rather than argued, by watching an untouched network's state move from 0 to 1 across a call made for a different one. Every consumer built as a loop over networks in the host therefore cannot fire a native rule without corrupting the networks its loop has not yet reached, and deciding when the engine should fire relative to such a loop is a question this arc never set out to answer and did not. Sixteen compiled functions were audited for removal and none was removed, because each still has at least one real consumer — the migrations are gated rather than unconditional, so both paths stay live. Across the arc's full seven consumers: two fully native, one partly, four still compiled.

3 The solver: what the oscillation turned out to be

Remark 3.1 (The oscillation was the update schedule, and the record found it). *(Record of 13 August 2026.)* What was observed is not in doubt and reproduces exactly. On a fixed eight-node draw with a fixed nonzero field per node and a fixed start, the relaxation returns different states at 400 and 401 iterations (opposite-parity gap 5.615746) and the same state at 400 and 402: a persistent period-2 cycle against a same-parity control. This paper's first revision reported that as a genuine binary between cohering and failing to cohere — an operation that either resolves or returns a determinate nothing — already present, unexploited, in the borrowed model's own mathematics, and so as an instance of the claim that such failure comes in one manner and not in kinds (Graphic Equalisation, Proposition 9.1).

It was the update schedule. The relaxation iterated $\sigma_i \leftarrow \tanh(\beta(\sum_j J_{ij}\sigma_j + h_i))$ fully synchronously — every node's update in a sweep reading the previous sweep's state — which is the textbook source of period-2 limit cycles in this family; a second, byte-identical copy of the same iteration was found in a sibling routine that the theoretical audit preceding the fix had not named. Blending each freshly computed value with its own prior value, at a factor registered in advance rather than tuned against these networks, makes the same draw converge cleanly (gap 0.000000); setting that factor to 1 reproduces the old numbers bit for bit on the same binary. The before and the after are one draw and one code change apart, not two experiments. Blending cannot move or remove a fixed point, only the trajectory to one, which is why nothing that already converged changed — and why the damped fixed point is a nearby genuine one rather than the average of the old cycle's two phases, exactly as a frustrated network with several fixed points should give.

The elimination is total across everything re-tested: 126 cells — the original 24-trial registered sweep, previously 6/12 and 8/12 oscillating by condition; a 30-cell sharpening previously up to 100% oscillating at $N = 32$; and a 72-cell extension past the registered scope to $N = 64$ and $\beta = 4.0$ — converge cleanly, every one, gap exactly zero. So the correction is precise, and it is the useful part of the whole arc: *the binary this account has elsewhere was not present in the borrowed mathematics; what was present was a Jacobi iteration.* What is *not* established is the general claim. No scheduling-independent failure to converge was found anywhere this search looked, and that is a bounded negative — complete graphs only, $\beta \leq 4.0$, $N \leq 64$, one initial condition per trial — rather than a proof that none exists. One later rung of the arc still calls

the binary real and confirmed again; it ran on the uncorrected solver, and re-running its sweep under the corrected one is named in the record as a follow-on it did not perform.

4 Performance: the complexity question is the wrong one

The natural objection to putting an Ising model anywhere is that its ground-state problem is hard. Barahona (1982) makes the sharp version available: the planar two-dimensional problem *without* an external field is solvable in polynomial time, while the same planar problem *with* a field, and the three-dimensional problem either way, is NP-hard. Every network here carries a nonzero field, so the hard case is the relevant one.

And it does not bind, because no exact ground state was being sought. The hosted solver computes a mean-field self-consistency fixed point — polynomial per iteration, unconditionally, at every network size. The actionable question was therefore never whether this scales exactly, but how good the approximation is and when it fails to converge. That question now has an answer, and the answer sharpens rather than removes the point: exact inference *is* now sought, where the treewidth permits it, and refused where it does not.

Remark 4.1 (The ladder, climbed)*. (Record of 14 August 2026.)* Mean field is the first and weakest step of a known ladder — mean field, then Bethe and loopy belief propagation, then Kikuchi and generalized belief propagation (Yedidia et al., 2005) — and it is weakest *precisely* in the regime of loops, strong local correlation and frustration, which is the regime the frustration and basin audits exist to detect. This paper's first revision observed that an entire coherent piece of statistical-mechanics vocabulary had been borrowed while the better half of the same literature sat one step away unused, with a caveat attached: Bethe and loopy propagation are themselves known to lose stable solutions in the same frustrated, field-bearing, low-temperature regime. The step was then taken, and the caveat was tested rather than repeated.

Loopy propagation, both variants. Max-product is the theoretically appropriate one here, because everything else in this toolbox is organised around a joint configuration rather than a per-node marginal; on the dense frustrated case it is never worse than sum-product across eight draws, and on every draw where it converges it ties or beats a twenty-restart search. Its convergence reliability then *inverts across topology*: it converges on 5 of 8 dense adversarial draws but only 4 of 15 real sparse ones — the opposite of the naive expectation that sparser is easier — and the cause was measured rather than left as an anomaly, a sustained period-4 message oscillation traced to one structural 3-cycle plus small-integer weights producing exact ties between candidate scores. Sum-product converges everywhere (15 of 15) and to worse joint states (mean frustration 2.142 against 1.693 for the restart search): the textbook warning that a marginal argmax is not a good joint configuration, measured here rather than cited. Neither variant reliably beats the restart search on the deployed networks, and both cost 20–200× its wall-clock per solve. So the caveat held where it was aimed, and was not the whole story: the worse convergence is on the sparse networks, where it predicted none.

Exact inference, where the width permits. Off that ladder rather than on it, exact minimum-energy inference by clique-tree message passing was built for low-treewidth networks. It agrees with literal brute-force enumeration on five real networks, float-tight, and cross-checks against the same energy function every other consumer uses; it refuses cleanly on the case whose treewidth is 15, in under two milliseconds and without mutating anything, rather than attempting an exponential search; and on a real cycle-topology network it finds a strictly lower-energy state (-5.743 against -5.102) than twenty random restarts. That last is the concrete content of what "exact" buys: a widened search finds the best of the basins its restarts land in, and this does not search basins at all.

Tractability, measured. The restart search exists because local-minimum risk was measured first, and not only on the adversarial case: 14 of 30 real deployed networks (46.7%) show nonzero spread across twenty starts, so a sparse near-tree topology does not by itself guarantee a benign

landscape. Treewidth was measured exactly on both graphs — 2 for the real template, 15 for the dense stress case — which is what decides where exact inference applies at all. And the real coupling matrix has effective rank ≈ 1.98 of 5: the couplings this model carries are substantially less independent than their count suggests.

5 Results

Remark 5.1 (The registered scaling prediction is reversed, and its magnitudes moved). *(Record of 13 August 2026.)* The primary registered prediction — that coarse quantization would yield fewer distinct basins than fine, with the gap widening in network size — *is not confirmed, and the measured direction is closer to reversed.* As first measured, coarse yields *more* distinct basins (6 against 4 at $N = 4$; 15 against 4 at $N = 8$), a higher entropy-weighted effective basin count, and higher mean frustration (2.10 against 1.55 at $N = 4$; 15.32 against 11.06 at $N = 8$); at $N = 16$ the two conditions converge to near-parity on every measure. The manipulation check holds throughout — coarse quantization does produce larger per-edge representational error at every N — so the intervention worked and the prediction about its consequence was wrong. That is the useful shape of a null and it is reported here rather than reinterpreted into a different prediction that would have survived.

Those figures were measured on the uncorrected solver of Remark 3.1. The reversal was subsequently replicated on five independent coupling draws per cell and the replicated cells re-run under the corrected one. *The direction holds*: the effective basin count is higher for coarse at both sizes (2.358 against 2.145 at $N = 4$; 3.616 against 2.088 at $N = 8$). *The magnitudes do not*: frustration falls by roughly 7–10× at $N = 4$ and 2–2.5× at $N = 8$, and the effective basin count roughly halves, because restarts landing on different phases of an oscillation had been counted as distinct basins. And one sub-claim does not survive at all — the concentration reading, that a coarse landscape is the less concentrated of the two, is gone at $N = 4$ (0.565 against 0.560) and reversed at $N = 8$ (0.405 against 0.575), which retires the gloss that coarse belief is the more easily knocked between alternatives. The single-draw frustration figures of the first paragraph were not themselves re-measured; the re-check ran on the replicated cells.

Two further results the arc produced, both on the uncorrected solver and both carrying that caveat. *A null*: there is no crossover network size anywhere in $\{4, 8, 12, 16, 24, 32\}$ under a criterion registered in advance, on either the plain or the structured coupling design. The criterion never fires, and the pattern does not approach it — the sign of the effective-basin gap flips three times across the six sizes. *And the strongest positive result in the whole arc*, which is not about bin count at all: the sharpest resolution effect measured anywhere is the presence or absence of a *neutral option.* A two-value vocabulary produces 1.6–2× the frustration of every vocabulary with three or more values, at every size tested (22.6 against a 13.9–15.3 band at $N = 8$; 122.9 against 48.4–61.9 at $N = 16$; 501.3 against 246.9–275.2 at $N = 32$), while three through eleven values show no reliable ordering among themselves. Under the corrected solver this is the one earlier finding that does not merely survive but *sharpens*, to 2.36× at $N = 8$ and 2.71× at $N = 16$. Removing the neutral option is not coarsening a scale; it is removing the representation's own way of declining to commit, and that is what the frustration is measuring.

Remark 5.2 (Resolution separates the basins, with a control that says so). *(Record of 13 August 2026.)* A second registered run, on a population of simulated residents each carrying its own network at its own representational resolution, confirmed both of its predictions.

The manipulation check is large: the coarse resident's mean rounding error is about 9.2 times the fine resident's (0.553 against 0.060). The primary prediction — deliberately not the naive basin-count intuition, that having already been falsified above on the same engine — is that under *identical* scenario evidence the coarse network relaxes to a positive-energy, highly frustrated basin (frustration 9.86, energy +5.19) while the fine one relaxes to a negative-energy, low-frustration basin (frustration 1.62, energy −1.77). It does.

A population-wide check across one hundred residents, reported as exploratory rather than pre-registered to the same standard, shows the same direction on average (mean frustration 3.29 in the low-resolution group against 1.38 in the high). *The control is what makes this worth reporting*: holding resolution fixed across the population collapses the trait-to-frustration correlation from -0.33 to -0.08, so resolution rather than some correlated trait is carrying it.

6 Falsifiability commitments

Each commitment below is marked [SHAPE] or [DETAIL]. A [DETAIL] failing is a revision: the account survives with that mechanism replaced. A [SHAPE] failing costs *range* rather than a mechanism, because what fails is something the substrate is claimed to require or to permit — so the account does not hold where it claimed to, which is the more serious of the two and is still a boundary rather than an annihilation.

Two of the five commitments made in this paper's first revision have *fired* against the record this paper reports from. Both are recorded as fired and replaced below, not quietly removed.

1. [SHAPE] **The dynamics are native, and the remaining boundary is scope.** Fails if the native rules' results diverge from the hosted solver's on the same network, or if the per-network scoping cannot be enforced in rule matching. *This replaces a commitment that fired.* The first revision committed that the migration was partial in the way it stated — "fails if the dynamics can be expressed in the existing term vocabulary without new term surface, that is, if the fifteen rules were inert for an ordinary reason after all." They can, and they were, one day after the stage that revision reported (Remark 2.2). The inertness diagnosis of Remark 2.1 stands; the conclusion drawn from it does not.

2. [DETAIL] **Resolution drives basin quality.** Fails if the frustration and energy separation survives holding resolution fixed, or if the population correlation does not collapse under that control.

3. [DETAIL] **The scaling prediction is reversed and not merely unconfirmed.** Fails if the basin counts recover the registered direction at $N = 4$ and $N = 8$ under a larger sample *on the corrected solver*. The first revision's form of this commitment — a larger sample on the uncorrected solver — has been run and did not fire: five independent draws per cell hold the direction. It is restated here against the corrected solver, on which the direction also holds (Remark 5.1) and the magnitudes do not.

4. [SHAPE] **Exact inference does not change what the separation says.** Fails if the frustration and energy separation of Remark 5.2 does not survive exact clique-tree inference on the treewidth-2 networks where it applies. The first revision's form of this — "fails if an exact solver at these network sizes changes the reported quantities materially" — had no instrument to fire against. It now has one (Remark 4.1), and running it against Remark 5.2 is the one thing on this list not yet done.

5. [DETAIL] **The oscillation is the solver's and not the model's.** Fails if a damped, asynchronous, or independently written solver reproduces a period-2 cycle on any of the 126 re-tested cells. *This replaces a commitment that fired.* The first revision committed, as [SHAPE], that "the oscillation is the model's and not the implementation's — fails if the two-cycle disappears under an equivalent solver written independently." It disappears under damping, on every cell re-tested (Remark 3.1). The replacement is [DETAIL] because what is now claimed is a fact about one iteration scheme, not a shape the account requires.

6. [DETAIL] **Max-product's convergence deficit on the sparse networks is the mechanism named.** Fails if breaking the tied candidate scores — perturbing the small-integer

weights — does not restore convergence on the networks where the period-4 message oscillation was measured.

References

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